

Image data compression
Microscopy, Modeling and Visualization
March 9 2006
D.P. Huijsmans



Fourier



Shannon



Huffman

Compression of Image data

Compact Representation of pixel positions and intensities

Matrix representation: pixel intensity matrix byte/pixel (& implicit pixel position information)

Still redundant representation because intensity values of neighbouring pixels are highly correlated (highly predictable)

More compact: uses less bytes $N_c < N_o$

Compression Ratio: $C = N_o/N_c$ preferably $C > 1.0$

Normalized Compression Ratio: $0 < N_c/N_o \leq 1.0$

Lossless and Lossy Compression

Lossless:

exact reconstruction of intensity pattern after decompression

Lossy:

approximate reconstruction of intensity pattern after decompression

Compression ratio can be very high for lossy compressed image:
No limit exists, but quality degrades for higher ratios

Theoretical Question:

what is the compression limit for lossless compression?

Measuring Redundancy: Entropy

Given the probability density function

$P(i)$ of a pixel state variable I , $0 \leq P(i) \leq 1$;

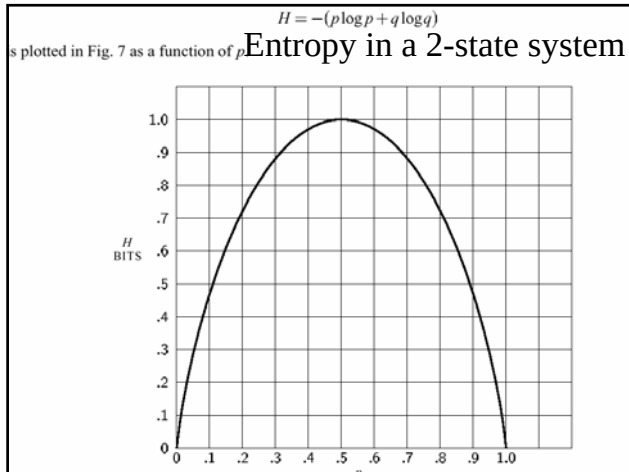
Entropy = $\sum (-P(i) \log_2(P(i)))$ bits

gives the minimal number of bits/pixel
needed to exactly reconstruct the pixel state i

Example of pixel state variable: intensity value

Probability density function of intensity: normalized histogram

N.B. Optimal compression with respect to one pixel state variable does not mean better compression rates are not possible according to another state descriptor
(for instance the pixel state describing the difference with the predicted intensity value from the left neighbour)



Entropy:
minimum and maximum value

Pixel intensity grey value 1 byte=8 bits/pixel I in [0,255]

Minimal chaos: all pixels in some state j ; $P(j)=1.0$, $P(i)=0.0$, $i \neq j$

$$E_{\min} = 0^2 \log(0) + \dots + 1^2 \log 1 = 0 \text{ bit/pixel}$$

$$E_{\max} = -n \cdot 1/n^2 \log(1/256) = -\log(2^{-8}) = 8 \text{ bit/pixel}$$

Variable length code words

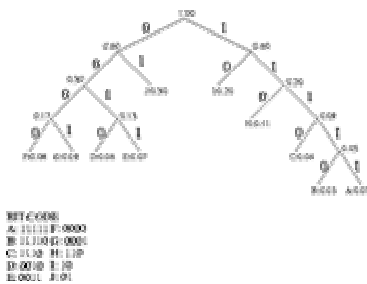
Bit-strings sorted according to length:

- 0,1,00,01,10,11,000,001,010,011,100,101,110,111,0000,0001,...
- How to distinguish code words without extra separator string?
- Uniquely decodable subsets:
- Start of next code word different from shorter code words:
- 0,1,00,01,10,11,000,001,010,011,100,101,110,111,0000,0001,...
- 0,-,-,-,10,-,-,-,-,-,-,-,-,-,-,-,110,111,-,-,-,-,-,-,-,-,-,
- 0,1,00,01,10,11,000,001,010,011,100,101,110,111,0000,0001,...
- --1--.-01-----000.001-----

Huffman code

- Uniquely decodable, variable length code
- Most efficient such code
- Method of code formation:
 - - count feature frequencies (normalized histogram)
 - - register feature-frequency connection
 - - while (more than 1 frequency) do
 - - - sort frequencies;
 - - - combine 2 lowest: register merged features in graph;
 - - enddo
- In merged feature graph: distribute 0 and 1 to splits
- Read code words by retracing graph from root to feature

Huffman code example



HUFFMAN-CODE

grys waarde	H-code	L_H	$L_H \cdot w(i)$	$w(i)$	$-\log_2 w(i)$	$-\log_2 w(i)$
0	0010	4	4/16	1/16	4	1/4
1	0011	4	4/16	1/16	4	1/4
2	010	3	3/8	1/8	3	3/8
3	010	2	1/4	1/4	2	1/2
4	11	2	1/4	1/4	2	1/2
5	011	3	3/8	1/8	3	3/8
6	0000	4	4/16	1/16	4	1/4
7	0001	4	4/16	1/16	4	1/4
3 bits/pixel			2,75			2,75
			R_H			E

$R_H = E$
 Maximaal haalbare resultaat

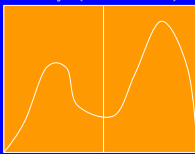
Prediction Model

In general Compression techniques are best applied to pixel error values

Error value = original pixel value – predicted value

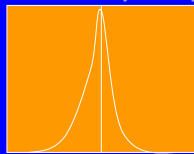
The better the prediction, the more peaked around zero the Probability distribution of the error; the lower the Entropy or bits/pixel needed

Prediction: Intensity= (maxint-minint)/2



Entropy=3.75 bits/pixel

Prediction: Intensity= Intensity of neighbour



Entropy=1.35 bits/pixel

JPEG

Lossy compression with quality factor $5 < q < 100$

Joint Photography Expert Group

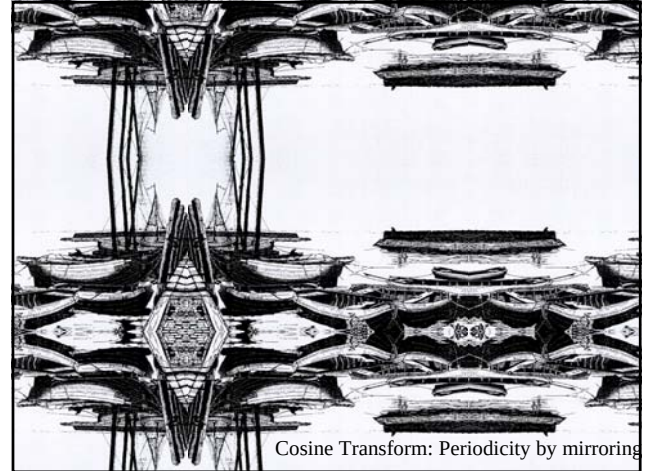
Coding with values from prediction model, so approximate

Prediction model: description with waveforms
Sum of waves of different wavelength (frequency)
In all directions, each with certain amplitude and phase

Model applied to small blocks (32x32 pixels) in image

In JPEG2000 wavelets are used instead of cosine transform

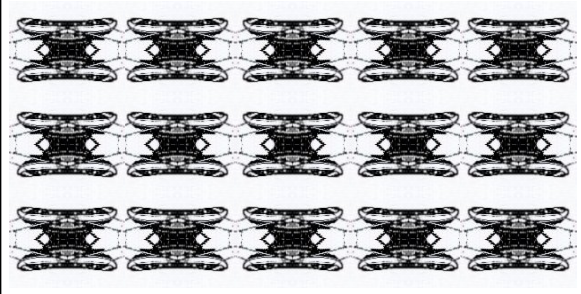
Fourier and Cosine describe periodic extensions of the image;
Wavelets describe damped periodic extensions of local areas



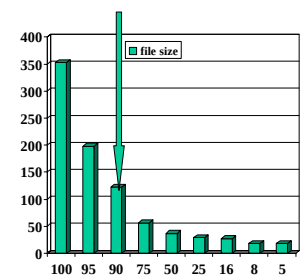
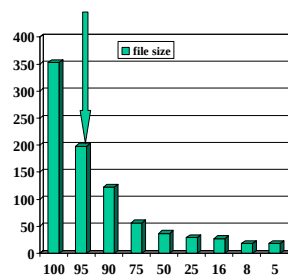
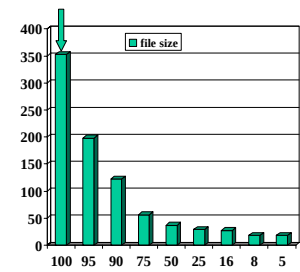
Fourier Transform: block area

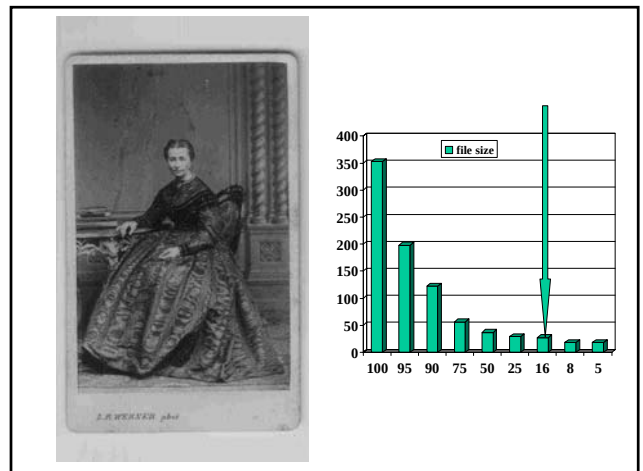
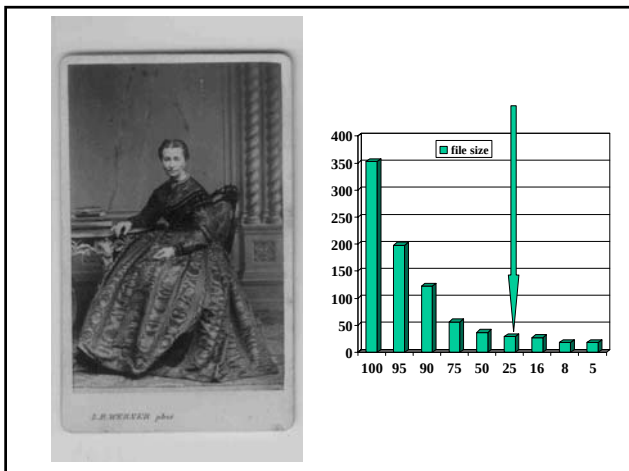
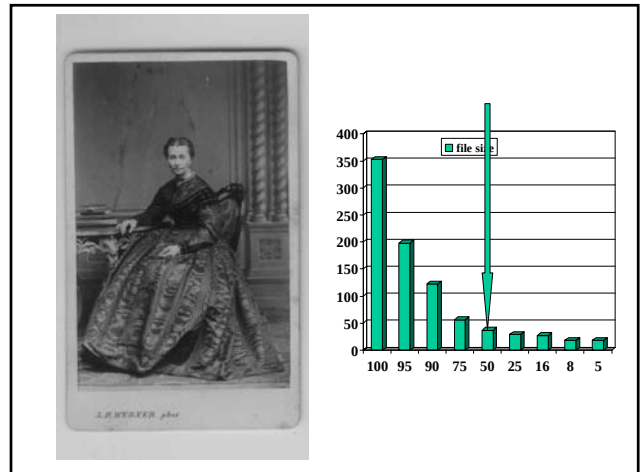
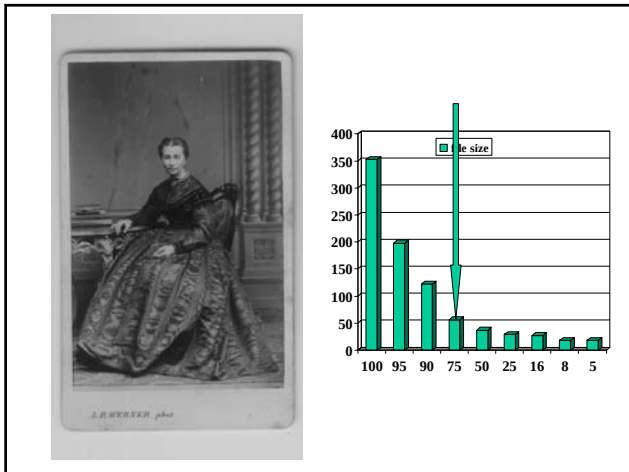


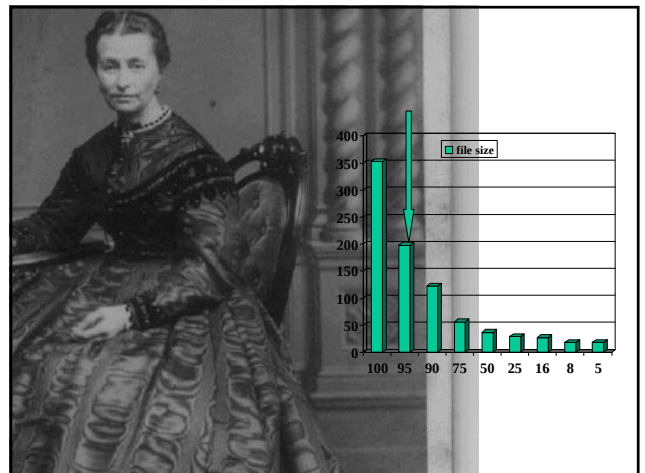
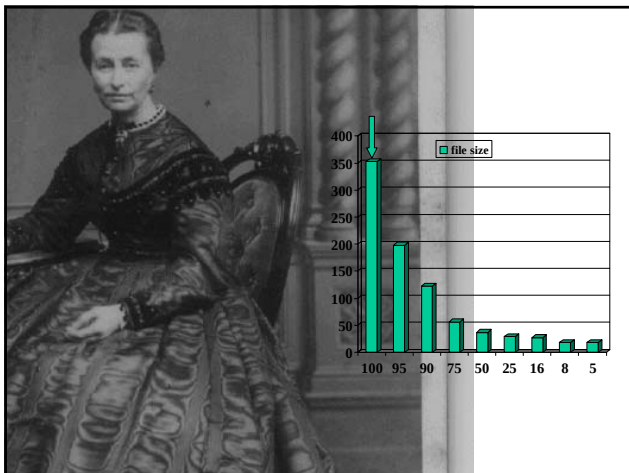
Cosine Transform: block area

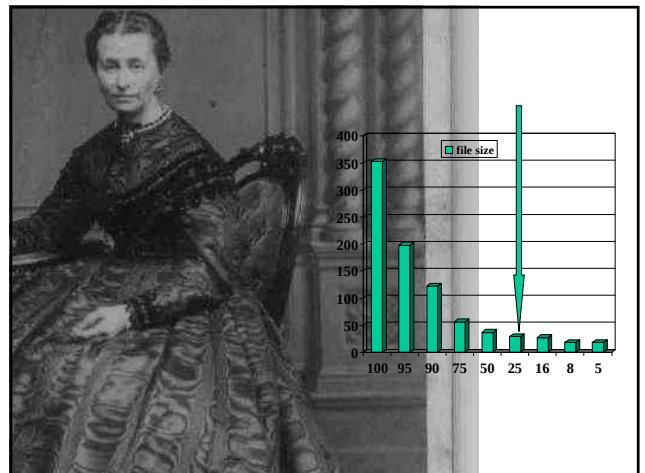
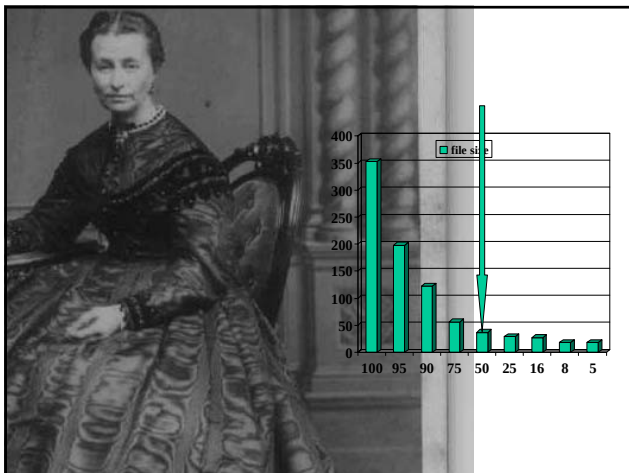
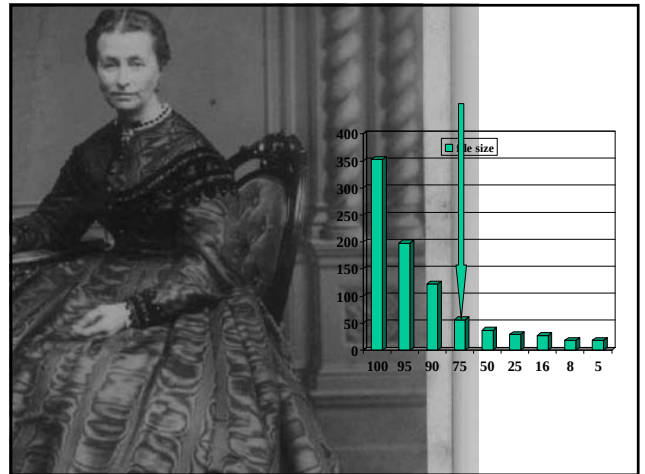
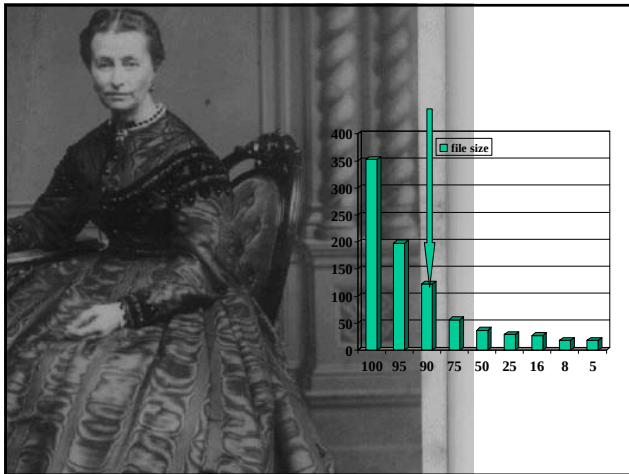


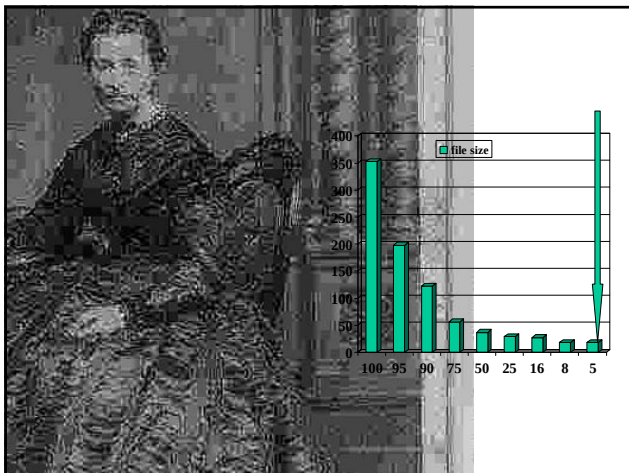
JPEG: blocking artifacts







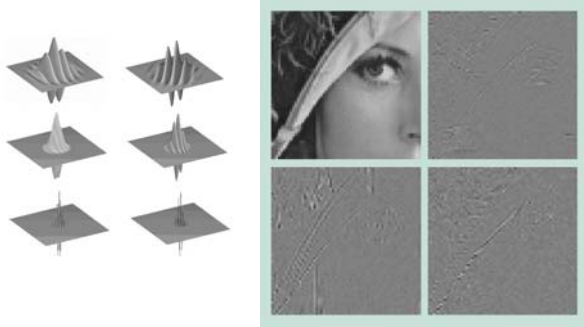




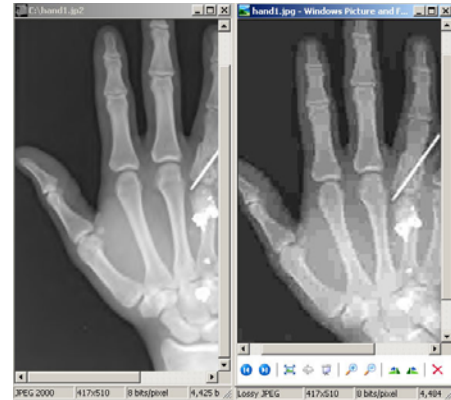
JPEG2000: wavelets

- Small damped wave packages:
- Localized (unlike Fourier or DCT)
- No blocking artifacts
- Less periodicity artifacts

wavelets



JPEG2000 versus JPEG



MPEG

Lossy compression of digital video frames (DVD)

Motion Picture Expert Group

Exploit correlation in space and time and between frames

MPEG2 DVD video coding (film on DVD)

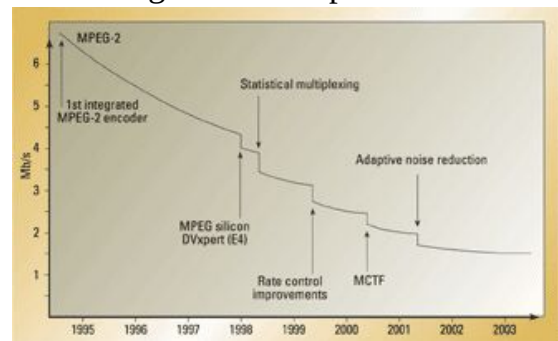
MPEG4 on the fly digitization

MPEG7 with feature vectors for image search

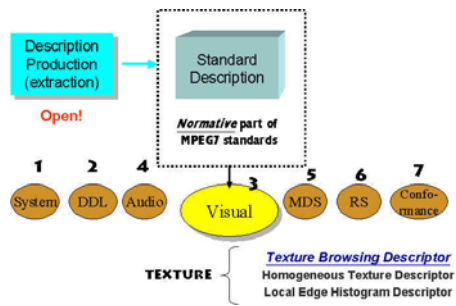
Latest developments: block motion between keyframes

MPEG-7: DVD will fit on CD !

MPEG2 compression progress Megabit stream per second

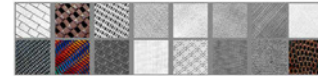


MPEG7

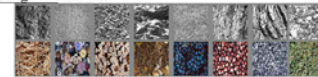


MPEG7 texture search

• Show textures which are not very coarse with high regularity and orientation around 30 degree



• Show irregular textures



• Show directional textures

